Introduction to Bootstrap

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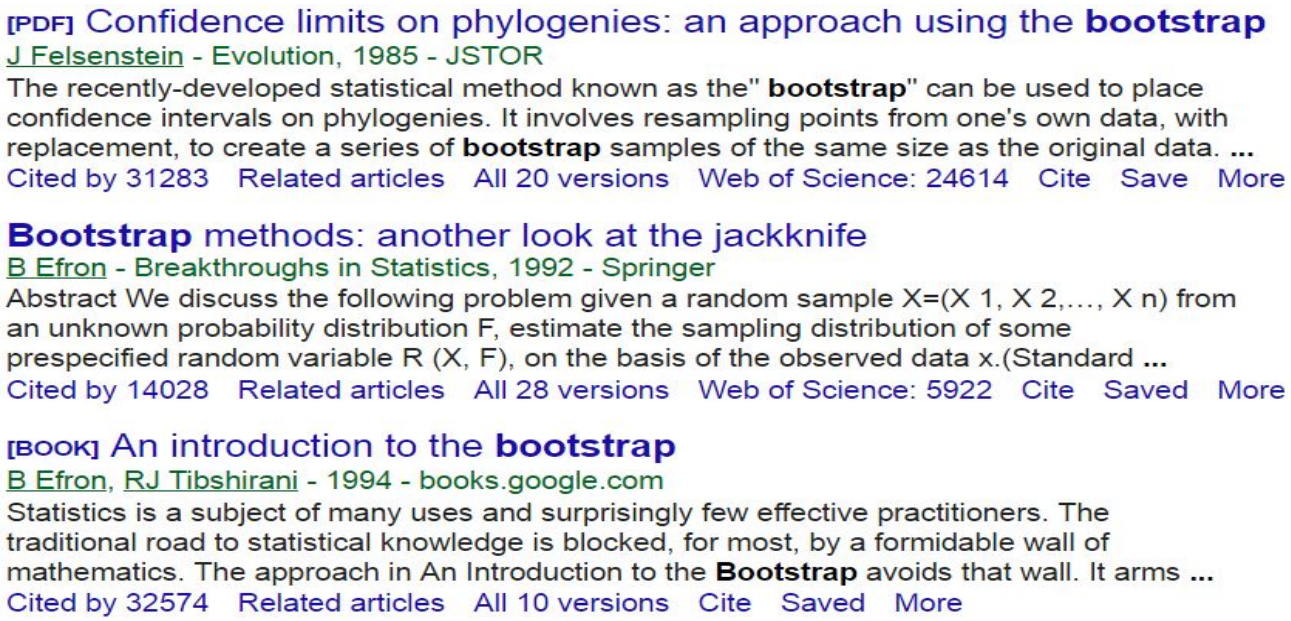
Cross-validation vs. the Bootstrap

Cross-validation: provides estimates of the (test) error

The Bootstrap: provides the (standard) error of estimates

What is Bootstrap?

- Wikipedia: group of metaphors which refer to a self-sustaining process that proceeds without external help.
- In Statistics, Bootstrap is a procedure that helps us get an idea about the distribution of a test statistic by resampling from the sample that produces the statistic.
- Introduced by Brad Efron in 1979 - spread like a bushfire in scientific disciplines within a couple of decades.


[PDF] Confidence limits on phylogenies: an approach using the **bootstrap**
J Felsenstein - Evolution, 1985 - JSTOR
The recently-developed statistical method known as the" **bootstrap**" can be used to place confidence intervals on phylogenies. It involves resampling points from one's own data, with replacement, to create a series of **bootstrap** samples of the same size as the original data. ...
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Bootstrap methods: another look at the jackknife
B Efron - Breakthroughs in Statistics, 1992 - Springer
Abstract We discuss the following problem given a random sample X=(X 1, X 2,..., X n) from an unknown probability distribution F, estimate the sampling distribution of some prespecified random variable R (X, F), on the basis of the observed data x.(Standard ...
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[BOOK] An introduction to the **bootstrap**
B Efron, RJ Tibshirani - 1994 - books.google.com
Statistics is a subject of many uses and surprisingly few effective practitioners. The traditional road to statistical knowledge is blocked, for most, by a formidable wall of mathematics. The approach in An Introduction to the **Bootstrap** avoids that wall. It arms ...
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Figure 1: One of the most-cited papers

Introduction & Motivation

- Goal: **Summarize** a sample based study and **generalize** the finding to the parent population, using sample statistics are sample mean, sample median, sample standard deviation etc.
- Sample mean will fluctuate from sample to sample and we would like to know **the magnitude of these fluctuations** around the corresponding population parameter in an overall sense.
- This is then used in assessing the Margin of Errors.
- The entire picture of all possible values of a sample statistics presented in the form of a probability distribution is called a **sampling distribution.**

Sampling distribution

- There is a plenty of theoretical knowledge of sampling distributions, which can be found in any text books of mathematical statistics.

Normal: $\mathcal{N}(\mu, \sigma^2)$	Sample mean $\bar{X}$ from samples of size n	$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
Bernoulli: Bern(p)	Sample proportion of "successful trials" $\bar{X}$	$n\bar{X} \sim \text{Bin}(n, p)$
Two independent normal populations:$\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$	Difference between sample means, $\bar{X}_1 - \bar{X}_2$	$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$

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- A general intuitive method applicable to *just about any kind of sample statistic* without the technical tedium has got its own special appeal.
- Bootstrap is such a method.

Motivation

- To understand bootstrap, suppose it were possible to draw repeated samples (of the same size) from the population of interest, a large number of times.
- For example, if you want to know about distribution of $\bar{X}$, where $X_1, \ldots, X_n$ are from $\mathcal{N}(\mu, \sigma^2)$, you can do so by drawing many independent samples of size n from $\mathcal{N}(\mu, \sigma^2)$, and see how $\bar{X}$ from different samples behave.

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- For any statistic, one would get a fairly good idea about the sampling distribution of a particular statistic from the collection of its values arising from these repeated samples.
- But, that does not make sense as it would be too expensive and defeat the purpose of a sample study. The purpose of a sample study is **to gather information cheaply in a timely fashion.**

Main idea

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- Sample with replacement: you have a box with n balls with n values, you 1) choose a ball at random 2) note it's number, 3) put it back, and 4) repeat. (this is called an **SRSWR:** Simple Random Sample With Replacement).

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- Sample with replacement: you have a box with n balls with n values, you 1) choose a ball at random 2) note it's number, 3) put it back, and 4) repeat. (this is called an **SRSWR:** Simple Random Sample With Replacement).
- The sample summary is then computed on each of the bootstrap samples (usually a few thousand).
- A histogram of the set of these computed values is referred to as the bootstrap distribution of the statistic.

R Demo

Let's try bootstrap for a simple problem:

https://dattahub.github.io/data-analytics/bootstrap_demo.html

Situation

1. You have observations from a completely unknown distribution, F.
2. If F is completely unknown, we cannot generate random numbers from the distribution.
3. $X_1, \ldots, X_n$ are the only realizations of F known to us.
4. The idea is to approximate F by $\hat{F}$ based on $X_1, \ldots, X_n$.
5. In other words: Inference about a population from sample data, (sample $\rightarrow$ population), can be modeled by resampling the sample data and performing inference about a sample from resampled data, (resampled $\rightarrow$ sample).

Algorithm

1. Data: $X_1, \ldots, X_n$. Test statistics t_n (e.g. $\bar{X}$)
2. Draw B samples $x_1^b, x_2^b, \ldots, x_n^b$ with replacement $b = 1, \ldots, B$ (there might be ties). This is called **a bootstrap sample**.
3. For each of the B samples calculate the statistic t_n^b.
4. Use these B realizations of t_n to compute whatever you want, e.g.
   - mean: $E(t_n) \approx 1/B \sum_b t_n^b$.
   - variance: $Var(t_n) \approx 1/(B-1) \sum_b (t_n^b - \bar{t}_n)^2$.
   - Empirical CDF $\hat{F}(t_n)$.

Bootstrap Uses

- The basic use of Bootstrap is for estimating the CDF $P_F(T \leq t)$.
- Over time, the bootstrap has found its use in estimating other quantities, e.g., $Var_F(T)$ or quantiles of T.
- The bootstrap is thus an omnibus mechanism for approximating sampling distributions or functionals of sampling distributions of statistics.
- Since frequentist inference is mostly about sampling distributions of suitable statistics, the bootstrap is viewed as an immensely useful and versatile tool, further popularized by its automatic nature.
- However, it is also frequently used in situations where it should not be used.
- We will first present some results on Bootstrap's accuracy and then show more examples (success and failure).

Empirical Distribution

- The EDF F_n is defined as $F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I\{X_i \leq x\}$.
- Glivenko-Cantelli Lemma: EDF converges to the true CDF, i.e. F_n converges to F as $n \to \infty$.
- Since the Bootstrap samples are resamples from the given sample, we can think of bootstrap samples as draws from the Empiricial Distribution Function.
- One way of describing Bootstrap is:

> The population is to the sample
> as
> the sample is to the bootstrap samples. (John Fox)

Bootstrap Distribution

- Suppose $X_1, \ldots, X_n \sim F$ and $T(X_1, X_2, \ldots, X_n, F)$ is a **functional**.
- For example, $T(X_1, X_2, \ldots, X_n, F) = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$ is an important functional.
- The formal definition of the bootstrap distribution of a functional is the following:

Definition
Let $X_1, \ldots, X_n \sim F$, and T be a given functional. The ordinary bootstrap distribution of T is defined as:

$$H_{boot}(x) = P_{F_n}(T(X_1^b, \ldots, X_n^b, F_n) \leq x)$$

where $X_1^b, \ldots, X_n^b$ is a resample from $X_1, \ldots, X_n$ or (equivalently) an iid sample of size n from the empirical CDF F_n.

Bootstrap Distribution

- When we talk make a probability statement under the empirical CDF P_{F_n}, such as $P_{F_n}(T \leq t)$ it corresponds to probability statements corresponding to all possible resamples with replacement from the original sample $(X_1, \ldots, X_n)$.
- How many possible resamples are there?
- Let me rephrase, you have a box with n balls. In how many ways you can select a sample of n balls if you draw with replacement? (Hint: it's not $n!$)
- Answer: n^n: Each of the n positions can be filled in n different ways.
- Since recalculating T from all n^n resamples is basically impossible unless n is very small, one uses a smaller number of B resamples and recalculates T only B times.
- This technique is generally called Monte Carlo, so the Bootstrap distribution $H_{Boot}(x)$ itself is estimated by a Monte Carlo, known as the bootstrap Monte Carlo.

Bootstrap Sample Size B

- so the final estimate for $P_F(T(X_1, X_2, \ldots, X_n, F_n) \leq x)$ absorbs errors from two sources:
  1. Pretending the Bootstrap samples to be bona fide resamples from F;
  2. estimating the true $H_{Boot}(x)$ by a Monte Carlo.
- By choosing B adequately large, the Monte Carlo error is generally ignored.
- The choice of B that would let one ignore the Monte Carlo error is a hard mathematical problem; Hall (1986, 1989a) are two key references.
- It is customary to choose $B \approx 300$ for variance estimation and a somewhat larger value for estimating quantiles. It is hard to give any general reliable prescriptions on B.

Does Bootstrap work?

- At first Bootstrap seems like a circular idea, and a bit too simple to actually work!
- What one means by the bootstrap working in a given situation depends on what one wants the bootstrap to do.
- For estimating the CDF of a statistic, one should want $H_{Boot}(x)$ to be numerically close to the true CDF $F(x)$ of T.
- So we need to know how to calculate "distance" between two CDF's $H_{Boot}(x)$ and $F(x)$.

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- So we need to know how to calculate "distance" between two CDF's $H_{Boot}(x)$ and $F(x)$.
- We have seen such a metric: the Kolmogorov divergence between two DF's F and G (recall K-S tests)

$$K(F, G) = \sup_{-\infty < x < \infty} |F(x) - G(x)|$$

Theory without Proof i

We can show that the Bootstrap is consistent!

Theorem

Suppose $X_1, \ldots, X_n \overset{IID}{\sim} F$ *and* X_i*'s have finite variance under* F*,* $E_F(X_1^2) < \infty$*. Consider the functional,* $T(X_1, \ldots, X_n, F) = \sqrt{n}(\bar{X} - \mu)$*. Then,* $K(F, H_{Boot}) \overset{a.s.}{\to} 0$ *as* $n \to \infty$*.*

- The notation $\overset{a.s.}{\to}$ stands for 'converges almost surely'.
- To say that the sequence X_n converges almost surely or almost everywhere or with probability 1 or strongly towards X means that

$$\Pr\left(\lim_{n\to\infty} X_n = X\right) = 1.$$

- The need for mentioning convergence a.s. is due to the fact that the bootstrap distribution H_{Boot} is a random CDF.
- That H_{Boot} is a random CDF has nothing to do with bootstrap Monte Carlo; it is a random CDF because as a function it depends on the original sample $(X_1, X_2, \ldots, X_n)$.

Theory without Proof ii

- The condition for consistency of Bootstrap is existence of variance, which is the same condition for which the central limit theorem holds.
- Rule of thumb: if a functional T admits a CLT, then the bootstrap would be at least weakly consistent for T.
- It turns out that, for certain types of statistics, the bootstrap approximation is (theoretically) more accurate than the approximation provided by the CLT.
- Because any normal distribution is symmetric, the CLT cannot capture information about the skewness in the finite sample distribution of T . The bootstrap approximation does so.
- So the bootstrap succeeds in correcting for skewness, just as an Edgeworth expansion would do. This is called Edgeworth correction by the bootstrap, and the property is called second-order accuracy of the bootstrap!

Confidence Intervals

- We haven't talked about confidence intervals before, and we won't go into details - but simply put, a 95% confidence interval is an interval calculated from the observed data, that will contain the true value of the parameter with a high probability.
- For example $[\bar{x} - 1.96 * \sigma / \sqrt{n}, \bar{x} + 1.96 * \sigma / \sqrt{n}]$ is 95% CI for normal mean.
- It means: "There is a 90% probability that the calculated confidence interval from some future experiment encompasses the true value of the population parameter".

Bootstrap's most elementary application: Confidence interval.
One produces a large number of "copies" of a sample statistic, computed from these phantom bootstrap samples. Then, a small percentage, say $100(\alpha/2)\%$ (usually $\alpha = 0.05$), is trimmed off from the lower as well as from the upper end of these numbers. The range of remaining $100(1-\alpha)\%$ values is declared as the confidence limits of the corresponding unknown population summary number of interest, with level of confidence $100(1-\alpha)\%$.

Reference

Statistical Methodology

Volume 20, September 2014, Pages 63–72

Bootstrap—An exploration

Jyotishka Datta[a], Jayanta K. Ghosh[a, b]

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Figure 2: Some new directions

More R examples and packages

https://dattahub.github.io/data-analytics/bootstrap_demo.html